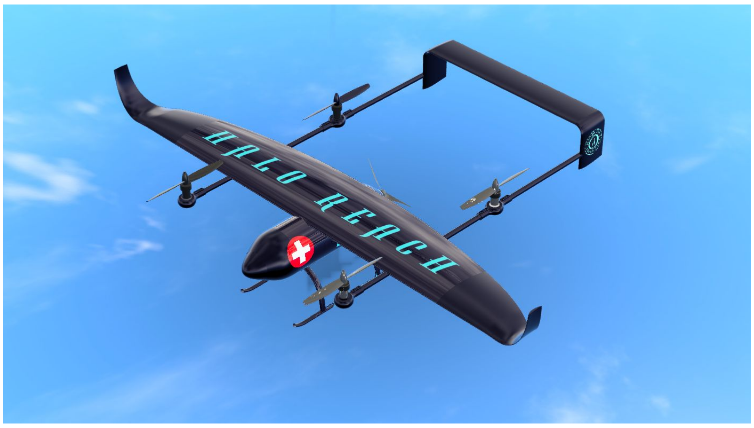
IMPERIAL

HALO REACH: Design and Optimisation of Winglets & Aerodynamic Tailplane Enhancements

UAS for Disaster Relief - AE3 2024



Student: Naboth Dirirsa
CID: 02026211

Supervisor: Kostas Steiros
Department: Department of Aeronautics
Course: MEng Aeronautical Engineering
Module: AERO60004 - Group Design Project
Date: 17/06/2024
Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	**Aerodynamics**	Alam, Shahriar	02037395
		Dirirsa, Naboth	**02026211**
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstarct

This report delves into the design and optimization of an aircraft winglet utilizing the Tornado Vortex Lattice Method (VLM) and STAR-CCM+ for Computational Fluid Dynamics (CFD) analysis. The primary objective is to enhance the aerodynamic efficiency and performance of the whole aircraft for long endurance. The study involves a comprehensive analysis of winglet configurations, leveraging Tornado VLM for initial design iterations, followed by detailed CFD simulations using STAR-CCM+ to validate and refine the designs. As a result, a blended winglet with a cant angle of 30° was designed which resulted in 15% improvement in both the endurance and range of the aircraft.

In addition to the winglet optimization, the report encompasses a CFD analysis of the tailplane, aimed at achieving aerodynamic improvements namely improving the geometry to reduce interference drag. This includes an assessment of blanketing effects, which are critical for understanding the interaction between the wing and tailplane, and for ensuring overall stability and control efficiency of the aircraft. The analysis found that the tailplane was completely blanketed after the wing had completely stalled.

Distribution of Work

To avoid repetition, we have endeavoured to keep each report as original as possible. However this means that some members of the team have been unable to present some of their work, due to some tasks being completed by multiple people. In order to try and give full credit to those who worked on topics not covered in their individual reports, a breakdown of work can be found in table 2. As some sections are highly interdependent, please refer to the relevant report for further information. We have tried to ensure that the individual work can be presented standalone however, information from other reports may add value.

Table 2. Separation of Work

Topic	People Involved	Person Presenting
Initial Concepts and Concept Selection	All	Parasdeep
Constraint Diagram	Max	Max
Wing Aerofoil Selection	Max	Max
Main Wing Design	Matthew and Max	Matthew
Analytical Drag Estimates	Max	Max
Winglet Design	Naboth	Naboth
Adkins and Leibeck Code	Parasdeep, Naboth, Shahriar	Parasdeep
Propellor Initial Thrust/Power analysis	Parasdeep	Parasdeep
Propeller Design	Parasdeep	Parasdeep
Propeller performance analysis	Parasdeep	Parasdeep
Rotor Initial Thrust/Power analysis	Shahriar	Shahriar
Rotor Design	Shahriar	Shahriar
Rotor Performance Analysis	Shahriar	Shahriar
CFD Setup and Validation for 2D Aerofoil	Matthew, Max	Max
CFD Setup and Validation For Stock Rotor	Matthew, Max	Matthew
Rotor CFD	Shahriar, Matthew	Shahriar
Propeller CFD	Matthew	Matthew
Full Wing CFD	Naboth	Naboth
Full Aircraft CFD	Max	Max
Blanketing Affects of Tailplane	Naboth	Naboth
Tailplane CFD	Naboth	Naboth

Contents

List of Figures

List of Tables

Nomenclature

α	Flow Angle of Attack
Æ	Aspect Ratio
$\frac{L}{D}$	Lift to Drag ratio
C_D	Drag Coefficient
C_L	Lift Coefficient
$C_{L_{design}}$	Design Lift Coefficient
VLM	Vortex Lattice Method

Part I
Winglet

1 Introduction

The UAV aircraft is expected to loiter for at least 10 hours and therefore all steps must be taken to ensure that its endurance and range are maximised. This would mean that an aircraft with high C_L/C_D is required at its $C_{L_{design}}$. Appropriate steps have already been taken to select an aerofoil that would maximise endurance and a semi-elliptical wing to ensure the best lift distribution [1].

Since induced drag is inversely proportional to the aspect ratio of the aircraft, making the wing longer and thinner means less induced drag on the wing overall [2]. However, this comes at the cost of increased bending moment on the wing due to the increase in the moment arm. Furthermore, the design requirement for our aircraft restricts the overall wing span from going beyond six meters and therefore, it is difficult to design a high aspect ratio wing. One way to mitigate this would be to have a wing that can fold during take-off and landing to meet the requirements and extend back to a high aspect ratio wing during the cruise segment of our mission. This would require a complex mechanism to be added to the wing to ensure it can operate reliably. This idea quickly becomes an issue as the weight of the wing would increase significantly as stronger material would need to be used to resist the increased bending moment and support the mechanism. Therefore, it was deemed unfeasible and disadvantageous to our weight-minimising and high stability goal.

Another way to reduce the induced drag without significantly increasing the aspect ratio and increasing the moment on the wing is using a wing tip device. Tip vortices on the wing have a lot of influence on the amount of induced drag. Experiments by NASA on the Boeing 707 with the use of winglets have shown a 6.5% reduction in fuel consumption [3]. Wingtip devices reduce the lift-induced drag by weakening the vortex strength at the end of the wing and therefore reducing downwash on the wing, resulting in overall lift and drag improvements. The type, shape and size of the wingtip device must be carefully selected and designed to ensure the drag reduction benefit outweighs the extra weight and bending moment on the wing resulting in an aircraft with longer endurance [4]. The following content of this chapter outlines the process taken to design and implement a blended winglet which resulted in 15% increase in endurance and 15% increase in the range of the UAS aircraft.

1.1 Exploring Wingtip Devices

As shown in Table 3, several wingtip devices were considered based on their aerodynamic advantages and other drawbacks that may undermine the benefits they provide as drag reduction tool.
The square wingtip improves effective $Æ$ slightly by pushing wingtip vortices outward, resulting in a minor reduction in lift-induced drag. Booster wingtips, both upturned and downturned, theoretically increase vortex separation but show negligible practical benefit. The Hoerner wingtip, named after aerodynamicist Sighard Hoerner, maintains a neutral impact on effective $Æ$ and enhances dihedral effect.

Table 3. Comparison of different wingtip devices based on aerodynamic effectiveness and other characteristics[5].

Wingtip Device	**Description**	**Aerodynamic Effectiveness ($\Delta \mathit{A\!\!R}$)**	**Notes**
Round Wingtip	Semi-circular edge extending from the leading to the trailing edge.	-0.19	Reduces effective aspect ratio (AR), less effective than predicted in practice.
Spherical Wingtip	Simple termination of the wing at the tip, usually a semi-circle along the tip chord.	-0.18	Causes wingtip vortex to roll up and inboard, reducing effective $\mathit{A\!\!R}$.
Square Wingtip	Sharp termination, forcing wingtip vortex to form on the outboard side.	+0.004	Increases effective $\mathit{A\!\!R}$ slightly, reduces lift-induced drag.
Upturned Booster Wingtip	Sharp termination flaring upward at the trailing edge.	0.00	Increases separation of wingtip vortices, but practical effectiveness is negligible.
Downturned Booster Wingtip	Inversion of the upturned style.	0.00	Perceived improvement in handling near stall, but limited practical benefit.
Hoerner Wingtip	Sharp edge separating upper and lower surfaces along the aft trailing edge.	0.00	Neutral impact on effective $\mathit{A\!\!R}$, increases dihedral effect.
Raked Wingtip	Extends the wingtip outward and backward, increasing span.	+0.5 to +1.5 (varies)	Increases effective $\mathit{A\!\!R}$ significantly, increases wing bending moments.
Endplate Wingtip	Vertical plates positioned at the tip of a lifting surface.	Up to +0.76 (for h/b = 0.4)	Reduces lift-induced drag, but adds parasitic drag and structural complexity.
Winglet	Vertical extension at the wingtip, usually curved.	Up to +1.5	Reduces lift-induced drag, increases wing bending moments, complex to manufacture.

(a) Blended Winglet **(b)** Fenced-tip **(c)** Split-tip winglet

The raked wingtip significantly increases the effective Æ but at the cost of increased wingspan and wing bending moments. Endplate wingtips, adding vertical plates at the wingtip, reduce lift-induced drag but introduce parasitic drag and structural complications. Lastly, winglets, a sophisticated evolution of endplates, effectively reduce lift-induced drag and are widely used in modern aviation, though they increase wing bending moments and manufacturing complexity. The winglet is selected as the final design because it works to overcome the constraint in aspect ratio and improve the L/D of the aircraft.

2 Design Methodology

2.1 Vortex Lattice Theory

The Vortex Lattice Method is based on Prandtl's lifting line theory. The main assumption is that the flow is incompressible and inviscid. The wing is modelled with a set of lifting panels with each panel containing a horse-shoe vortex. The Tornado VLM is different here, where the horseshoe vortex has been replaced by a vortex-sling arrangement shown in Figure 3. Although the physics is the same, the vortex-sling configuration have legs that are flexible and it includes seven vortices of equal strength instead of the three [6]. This extra flexibility feature allows it to capture the wake coming off the trailing edge better in different flight conditions which means the aerodynamic coefficients are slightly more accurate.

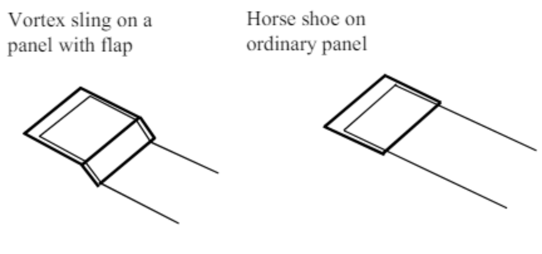


Figure 3. Tornado's seven-segment vortex sling vs the horse shoe[6].

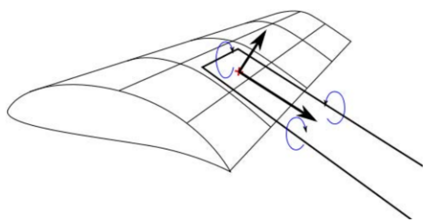


Figure 4. Placement of the horseshoe vortex on each panel of the wing [7]

As shown in Figure 4, each panel will contain a bound vortex placed at the quarter chord point. Variation of lift can then be modelled as step changes from one panel to another. It is set up in such a way that each panel would affect all the other panels and the velocity at the control points on each panel is calculated using the Biot-Savart law. The lift created by each panel is calculated and summed for the whole wing as shown by Equation 1 [8].

$$L_i = \rho V_\infty \Gamma_i 2k \qquad\qquad L_{\text{wing}} = \sum_{i=1}^{N} L_i \tag{1}$$

By calculating the lift component that acts parallel to the panel, the lift induced drag can be calculated as shown in Equation 2. Note that α_i represents the angle at which the lift vector has rotated such that it is parallel to the flow. By summing up all the induced drag on each panel, the lift-induced drag for the whole wing can be calculated as shown in Equation 2 [8].

$$D_i = \rho V_\infty \Gamma_i 2k sin(\alpha_i) \qquad\qquad D_{\text{wing}} = \sum_{i=1}^{N} D_i \tag{2}$$

Since the aircraft is analysed in the loiter phase of the mission the Mach number is well below 0.3 at which compressibility effects can be ignored. Furthermore, winglets are purposed to reduce lift-induced drag which is independent of viscosity and so the inviscid assumption is not a problem.

2.2 Optmisation of winglet with Tornado VLM

2.2.1 Tornado VLM validation with wind tunnel data

To ensure the VLM method being used is valid for the analysis of different winglet configurations, research has been conducted. Research in an academic report shows that VLM predicts the C_L/C_D values poorly as viscosity effects are not captured. However, the change in C_L/C_D values due to different wing configurations is captured quite accurately when compared to wind tunnel data. Figure 17 in Appendix A shows a study conducted on winglets at different cant angles using both Tornado VLM and wind tunnel[7]. The study shows that Torndo VLM accurately predicts the effect of winglets on C_L/C_D in terms of improvements. However, it greatly over-predicts the actual values of C_L/C_D due to the negligence of viscosity.

It is therefore reasonable to assume that Tornado VLM can capture the effectiveness of winglets and, hence can be used to optimise the geometrical configuration of the winglet. Furthermore, the vortex lattice method can accurately capture induced drag which is what the winglet aims to reduce so it can be concluded it is sufficient enough to use for the preliminary design of the winglets.

2.2.2 Set up on Tornado VLM

Tornado VLM is an open-source software package that runs in MATLAB [6].Figure 5 and 6 show how the mesh is set up in Tornado VLM using a linear panel distribution. Each square mesh represents a panel and therefore the finer the mesh the more accurate the results.

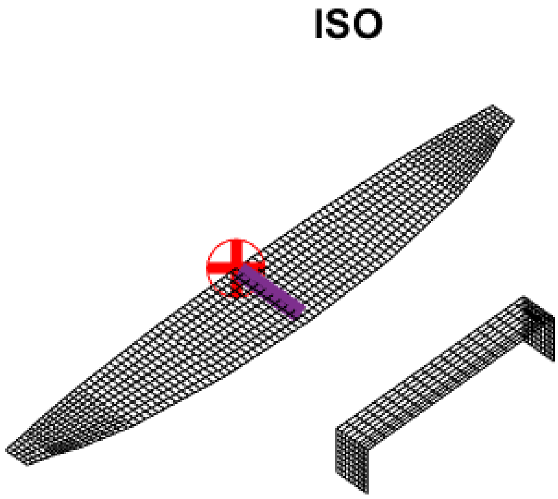


Figure 5. Clean Wing with tailplane set up on Tornado VLM

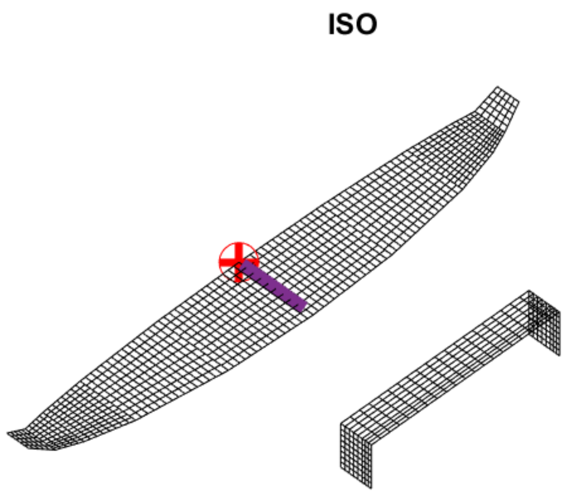


Figure 6. Wing-winglet and tailplane set up on Tornado VLM

To ensure that the elliptical distribution of the wing is not affected largely by the presence of winglets, only the last $0.25m$ (approximately 8.3% span) of the wing was used for winglet design. The design process consisted of changing the cant angle of the winglet and then calculating the effects on the lift and drag of the overall wing. Note that the cant angle is defined as the angle between the horizontal axis of the wing and the winglet.

All the studies on the effects of the winglet are done at the loiter speed of 27.6 m/s, an altitude of 500m and $\alpha = 3.6°$ as this is the segment of the mission in which optimisation of endurance is needed most.
Several wing geometries were created with winglets at different cant angles and each time the simulation was run with the same mesh resolution to ensure the results are not affected by it. Note that Tornado VLM has an inbuilt mesh convergence study which was run before every simulation to ensure accuracy. Each time the simulation was run, the values of lift, drag and their coefficients were recorded.

2.2.3 Tornado VLM Results and Discussions

The values of C_L/C_D were calculated at the different cant angles and the percentage increase from the clean wing configuration is plotted as seen in Figure 7. Along with the percentage increase in C_L/C_D the percentage increase in aspect ratio is also plotted in Figure 7. This is to ensure that the structural impact of the winglets is crudely considered as the increase in aspect ratio would generally increase the bending moment on the root of the wing. Note that the change in aspect ratio is calculated using an empirical relation for a winglet which has a maximum $\Delta AR = 1.5$ as shown in Equation 3 [5].

$$\Delta AR = 1.9\left(\frac{h}{b}\right)AR \tag{3}$$

Where h is the height of the winglet and b is the wingspan [5].

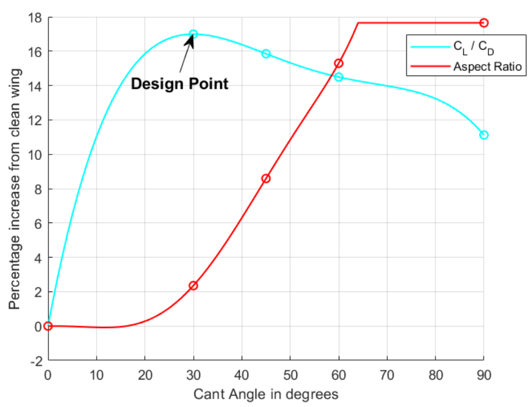


Figure 7. C_L/C_D and AR increase at different cant angles

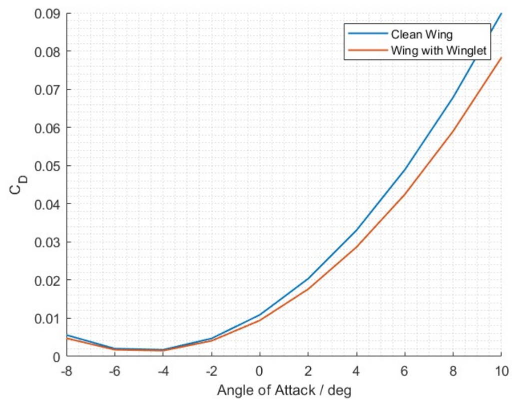


Figure 8. Variation of C_D with respect to angle of attack for clean and winglet wing confugration

Figure 9. Overall caption for the two images

It is evident that the maximum increase in the C_L/C_D occurs at a cant angle of 30° at which it increases by 17% and the corresponding increase in the aspect ratio is 2%. After the cant angle of 30° was chosen the simulation on Tornado VLM was then run at different angles of attack for both the clean wings and the wing with the winglet. Figure 8 shows the difference in the drag coefficient(C_D) before and after adding the winglets at various angles of attack. From this result, it is evident that the winglet becomes more effective at reducing drag at higher angles of attack at which the lift-induced drag is dominant. This also shows that the aircraft would be able to operate a lot more efficiently when climbing and descending. Furthermore, it shows that the addition of winglets is very likely to delay stall of the wing.

2.3 Tornado VLM validation with CFD in Star CCM+

Since the flow characteristics in vortex lattice method used in Tornado may be simplified, it is important to ensure the validity of the results by comparing it to other solvers. An industry-level CFD software called Star-CCM+ was used due to it's historical accuracy in predicting flow behavior, especially at low Reynolds numbers.

2.3.1 Computational Fluid Dynamics in Star CCM+

The model of the wing with and without winglet was set up on Star CCM+ using a virtual wind tunnel, where all the side walls apart from the symmetry wall are set with a slip condition to ensure the boundary layer of the walls does not interact with the rest of the flow. The physics model used in the simulation is outlined below[9].

Physics Model:

- **Flow type:** Segregated solver is used as it is ideal for incompressible low-speed flow. Therefore it assumes constant density and is based on an iterative procedure to solve the pressure-velocity coupling using a simple algorithm.

- **RANS turbulence model:** K-Omega model is used as it provides a good compromise between computational cost, robustness and accuracy. They also work very well for low Reynolds number regions and with flows with adverse pressure gradients and separation. It is also important to note that this model neglects diffusion of shear stress, which should have minimal impact on the study as it is quite focused on induced drag This model is often used in the aerospace industry as it is deemed reliable.

- Overall, the fluid is set up as steady and constant density gas which is quite reasonable due to the very low mach number of the flow .

The pressure distribution over the wing with and without the winglets is also visualised in Star CCM+ as shown in Figure 10 and 11. It is evident that the addition of the winglets improves the pressure distribution on top of the wing ensuring that the lower pressure is maintained for longer chord-wise. This is shown with a bigger blue region in Figure 11 compared to Figure 10 which further supports the ability of winglets to encourage a more two-dimensional flow and reduce the tip vortex strength(which increases the pressure on top of the wing indicated by the red colour in Figure 10)

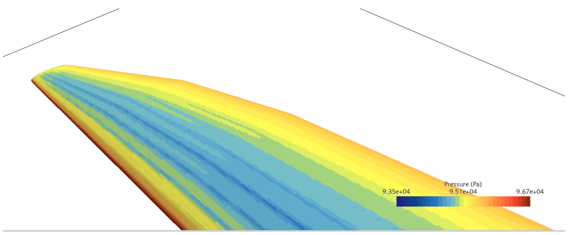


Figure 10. Clean wing pressure distribution

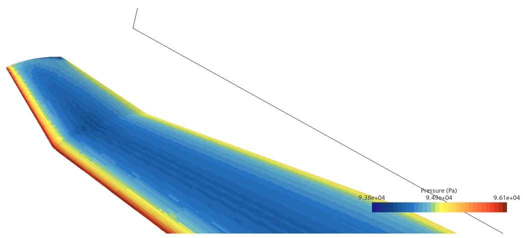


Figure 11. Wing-winglet pressure distribution

2.3.2 Mesh Convergence Study

To ensure the validity of the CFD simulation in Star-CCM+, a mesh convergence study was conducted for the aircraft wing by assessing the effects of changing the mesh on the value of drag. Since the geometry was quite large the mesh refinement was focused on the wing and the symmetry plane wall closer to the wing which is shown in Figure 18 in Appendix B. This is because the rest of the flow field was less important for the purpose of this study. The base size for the mesh was varied from $1m$ to $0.2m$, below the base size of $0.4m$ the change in Drag value was below 2% which was deemed sufficient as reducing it further increased computation cost significantly [9]. The mesh convergence graph can be seen in Appendix B, Figure 19

2.4 Results and Comparison

As expected, the values of C_L/C_D obtained from CFD are lower than the values from VLM, this is due to the effects of viscosity which is taken into account in CFD. The added wetted area of the winglet increases the skin friction drag which increases the drag slightly which accounts for the 2% difference between CFD results and VLM results.

$$\text{Range} = \frac{\eta}{cg}\frac{C_L}{C_D}\ln\left(\frac{W_0}{W_1}\right) \qquad (4)$$

$$\text{Endurance} = \frac{\eta}{cg}\frac{1}{V}\frac{C_L}{C_D}\ln\left|\frac{W_0}{W_1}\right| \qquad (5)$$

The range and endurance of a proppeler aircraft can be approximated using Breguet's Equations 2.4 and 2.4 respectively which relies upon the assumption that "the rate of aircraft weight reduction " is equal to "the rate of fuel weight burned". Although the key takeaway from the two equations is that C_L/C_D is directly proportional to both the range and endurance of propeller-based aircraft, it may be beneficial to note that the equations are also a function of propeller efficiency (η), specific fuel consumption (c) and the fuel fraction(W_0/W_1) [10].

Table 4. Comparison of CFD and Tornado VLM

	CFD	Tornado VLM
CL/CD	23.8 to 27.4	32 to 37.5
Total Drag Reduction	8%	9%
Endurance	15%	17%
Range	15%	17%

The final results for a 30° angled winglet show a significant improvement in the endurance of the aircraft at the loiter segment of the mission with a significant increase of at least 15%.

3 Effects of winglets on wing structure

To quantify the effect of winglets on the structure and therefore weight of the wing, the bending moment with and without winglets on the wing was calculated [11]. The percentage increase due to the addition of winglet on the wing is shown in Figure 12, which shows a 2.2%

increase in bending moment at the root of the wing. The bending moment was calculated using Tornado VLM which calculates the shear forces due to the aerodynamic forces and approximated weight of the wing through spanwise discretisation, hence calculating the bending moment from it [12].

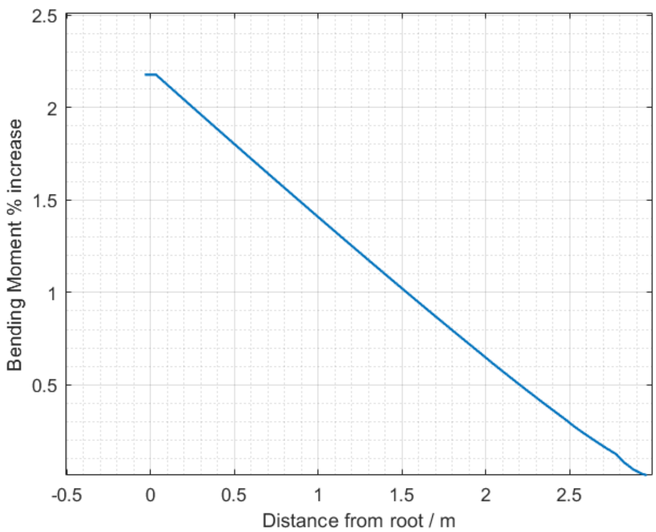


Figure 12. Percentage increase in bending Moment along the wing due to addition of the winglet

For simple analysis, it is sensible to assume that the bending moment of the wing half is reacted entirely by the spars and skin-stringer panels. Therefore as outlined in [5], the relationship in the bending moment at the root (M), the spanwise location of the aerodynamic centre (Y_{MGC}), ultimate structural load factor(n_{ult}) and the weight of the spars (W) shown in Equation 6.

$$M = \frac{n_{ult} W}{2} \times Y_{MGC} \tag{6}$$

From this, it can be concluded that the weight of the spars is directly proportional to the root bending moment: a 2.2% increase in bending moment corresponds to a 2.2% increase in spar weight. Similarly, the skin stringer panels can be idealised as a sheet of metal with an effective thickness T ($T = t + \frac{A_s}{b}$, where A_s is the cross-sectional area of the stringers and b is the chord-wise distance between each stringer). The relationship between longitudinal stress σ_z, Compressive load N and effective thickness is shown in Equation 7 [12].

$$\sigma_z = \frac{N}{T} \tag{7}$$

If a negligible change in the moment of inertia of the structure due to the addition of winglets, the stress can be assumed to be directly proportional to the bending moment on the wing. Therefore the effective thickness of the skin-stringer panel would need to increase proportionally to withstand the higher stress level. Increasing the effective thickness means increasing the combined cross-sectional area of the skin-stringer panel. Increasing the cross-sectional area by 2.2% would increase the volume by the same amount so the weight can also be assumed to increase by the same amount.

Since the weight of the spars and skin-stringer panels accounts for over 91% of the wing's structural weight [5], we can assume a 2% increase in the wing structural weight due to the addition of winglets. The final values on structural weight of the wing is outlined by Stephen's report[13].

4 Effects of winglets on Aircraft Stability

It is very important to consider the effects of winglets on both static and dynamic stability of the aircraft to ensure that the improved performance does not come at the cost of less stable aircraft. Especially, since the aircraft is very likely to operate in harsh environments where the conditions are unpredictable, it is crucial to ensure the aircraft's stability is robust and reliable. For the purposes of this study, a model of the wing (with and without winglets) and tailplane have been set up in Tornado VLM to estimate the stability derivatives. For simplicity, only stability modes have been analysed quantitively whilst some are discussed qualitatively using historical data.

4.1 Effects of Winglets on lateral and longitudinal stability

4.1.1 Lateral Static Stability

Lateral static stability is concerned with the tendency of the aircraft to correct its roll angle when perturbed in trimmed flight. In order to fully grasp the effects of winglets on the lateral stability, the stability derivatives, C_{l_β} , C_{n_β}, C_{y_β}, C_{l_p} and C_{n_r} are calculated before and after adding winglets [14]. As shown in Table 5, there is an increase in C_{n_β} which defines the yaw stiffness of the aircraft at all the three angles of attack. C_{n_β} increasing therefore corresponds to a restoring yawing moment increase for the aircraft which improves the airworthiness of the aircraft which is crucial, especially in the harsh wind conditions it is likely to be exposed to. It is also evident that there is a negative increase in the derivatives, C_{l_β}, C_{y_β} and C_{n_r}, C_{l_p} which are all stabilising to the lateral static stability of the aircraft.

Table 5. Lateral Stability Derivatives at Different Angles of Attack

Stability Derivative	Percentage Change		
	$\alpha = -5$	$\alpha = 0$	$\alpha = 5$
$C_{l\beta}$ (Roll Moment due to Sideslip)	-180	-18	-2
C_{l_p} (Roll Damping)	-2.6	-1.6	-0.2
$C_{n\beta}$ (Yaw Moment due to Sideslip)	8.2	0.35	1.1
C_{y_β} (Side Force due to Sideslip)	-6.53	-6.81	-7.28
C_{n_r} (Yaw Damping)	-0.25	-2.23	-2.55

Note: the final values of Lateral static stability derivatives are outlined in Ruairidh's report , which discusses the conditions of the lateral stability derivatives needed for aircraft stability[15].

4.1.2 Longitudinal Static Stability

Longitudinal static stability is concerned with the ability of the aircraft to rueturn to a trimmed pitch angle after perturbation. With winglets the aircraft reaches the required lift coefficient at a smaller angle of attack than without winglets, this means that adding winglets actually moves the aerodynamic centre aft and therefore affects the static margin. This would of course cause the C_{m_α} to increase negatively. It is important to note is quite useful for our aircraft as it shows that there would be a big improvement in drag performance at low speed as the aircraft would be able to loiter at a lower angle of attack than without winglets. Adding the winglet also results in an increase in the C_{L_α}. Equation 8 shows the relationship between C_{L_α} and C_{m_α} and the static margin [14]:

$$C_{m_\alpha} = C_{L_\alpha}(\bar{x}_{cg} - \bar{x}_{ac}) \tag{8}$$

The increase in static margin would mean z more negative horizontal stabiliser setting angle is required to trim the aircraft in cruise and loiter but the centre of gravity of the aircraft can be moved aft to mitigate this. Overall, it is evident to note that the increase in the static margin corresponds to an increased longitudinal stability of the aircraft.

4.2 Dynamic stability

Studies on KC-135A aircraft show that the addition of winglets on the wing has a negligible effect on the phugoid mode. Furthermore, it has also been shown that they are stabilising to the dutch roll, spiral and rolling characteristic modes shown by significant reduction time to half amplitudes and an increase in damping ratio. However, they are also known to slightly increase the relative amplitudes of short period oscillations. In summary, winglets continue to prove their positive contribution to aircraft's dynamic stability[7].

4.3 Conclusion

Overall, the winglets at a cant angle of 30° a have a significant effect on increasing endurance by at least 15% with a minimal weight-increasing penalty of 2%. Therefore it was successfully used to overcome the constraints in increasing the wingspan to reduce induced drag. Furthermore, winglets proved useful in improving both lateral and longitudinal static stability with the exception of a small trim penalty. Finally, winglets have been also shown to improve dynamic stability with a slight negative impact on SPO (Short period oscillations). Note that although all the study was done on a canted winglet for the sake of simplicity, the final design will be a blended winglet which would perform slightly better than the canted winglet [5].

4.4 Future Improvements

Due to the aircraft's low mach number operation, the sweep and twist of the winglet was not considered largely, furthermore, it was neglected due to its design complexity and manufacturability. It is also important to note that the selection process of the wingtip device was qualitative as time didn't allow for a thorough analytical comparison. Furthermore, the study on the stability derivatives did not consider the effects of the fuselage, booms and the propellers which may have impacted the results slightly. In future iterations, more sophisticated software such as FLEXSTAB can be used to captivate overall stability impact on the whole aircraft [7].

Part II
Tailplane Optmisation

1 Tail-plane Aerodynamic Analysis

This section of the report focuses on tail-pane aerodynamic analysis in Star-CCM+ and Tornado VLM to obtain accurate results for drag and reduce it. Furthermore the blanketing effect of the wing on the tail was also assessed at high angles of attack to ensure that the aircraft is able to recover from stall.

1.1 Interferance Drag Reduction

Due to the nature of our aircraft configuration in which a large empennage is needed to produce a sufficient moment for stability. Due to the large surface are of the tail and the double T-configuration, a lot of parasitic drag is created. Therefore, it was vital to ensure that all necessary steps are taken to reduce it's drag. Due to the sharp corner of the joint between vertical and horizontal tail-plane, a lot of drag is created as the flow between the two lifting surface interact. To reduce the interference drag the sharp corner was curved ensuring that the area required for rudder and elevator were not affected [16]. Figure 13 shows the original tail-plane design whilst Figure 14 shows the improved tail plane design mesh. The two tail-plane configurations were set up in CFD with similar physics model used for the wing and the magnitude of drag was noted at the cruise speed.

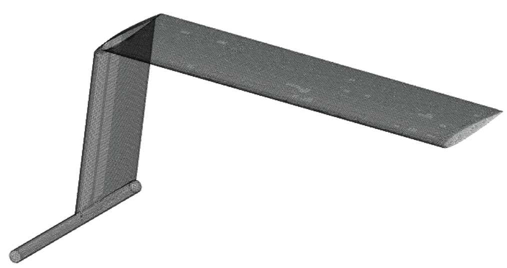


Figure 13. Sharp Corner Tailplane

Figure 14. Curved Corner Tailplane

The same model was also set up in Tornado VLM as a form of verification. Results from both VLM method and CFD method show a notable decrease in drag at the cruise speed which would contribute to ensuring the aerodynamic efficiency of the overall aircraft. The final results are outlined in Table 6. The drag reduced by 7.5% from on CFD and 8.2% on Tornado VLM, the difference in drag reduction is possibly due to the effects of viscosity which was not considered in VLM. However, both results show significant reduction in drag with very minimal impact on Lift which were both very close to zero due to the symmetrical nature of the aerofoil.

1.2 Blanketing effect analysis with Star-CCM+

The reasoning behind tail configuration choice has already been discussed by Ruairidh [15]. It is important to note that the high tail configuration's has the disadvantage of being in the wake of the wing at a high angle of attack which diminishes the elevator's authority

Table 6. Comparison of Drag with CFD and Tornado VLM

	CFD	Tornado VLM
Drag / N (Sharp - Curved corner)	9.94 - 9.24	10.94 - 10.11
Total Drag Reduction	7.5%	8.2%

to recover the aircraft from stall. Therefore a 2-D model of the tail and the wing was set up in Star-CCM at the wing's stall angle of attack to analyse the blanketing. As shown in Figure 15, the mesh on Star-CCM+ was set up such that the area between the wing and the horizontal tail had a finer mesh than the surrounding area to ensure the wake of the wing is captured correctly. Note that mesh convergence study similar to Section 2.3.2 was done and a base size of $0.01m$ was chosen for the final mesh

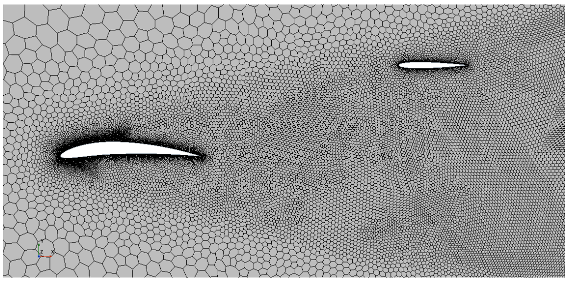


Figure 15. Mesh Set up on CFD

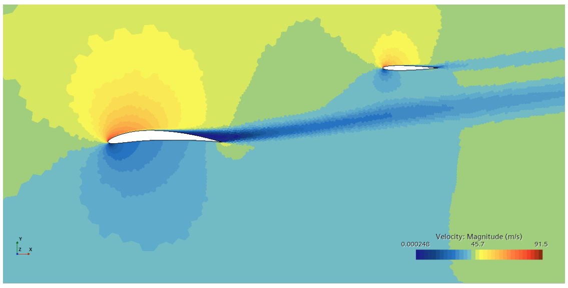


Figure 16. Velocity contour on CFD

Figure 16 show the velocity field and hence the wake of the wing at the angle of attack of 13.5 degrees. From the velocity field it is evident that the horizontal tailplane and therefore the elevator is not blanketed when the wing is on the point of stall. A further study was also done for $\alpha = 20$ degrees which shows the majority of the horizontal tail being blanketed. This is shown in Appendix C, Figure 20. This extra study was done to show the velocity profile that would represent a blanketed tailplane at which the authority of the elevator has completely diminished. Fortunately, this particular UAV is able to use the vertical rotor to recover in such extreme events.

1.3 Conclusion and Future Improvements

Interference drag plays a significant role when designing long endurance aircraft as it evidently accounts for a minimum of 7.5% overall drag as summarised in Table 6. However, there are lot's of joints on the aircraft at which interference drag can be reduced such as the joint between the wing and the fuselage; skid and the fuselage, and other joint interfaces. It is also evident that the horizontal tail is completely free from the blanketing effect of the wing. In the future, the blanketing effect of the pusher propeller, wing and tailplane can be analysed further. However, this was not possible at this stage of the design due to its computational complexity.

References

[1] Matthew M Homa. Halo reach: Wing design and computational propeller analysis. GDP report 02038403, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[2] E.L. Houghton ...et al. *Aerodynamics for Engineering Students*. Elsevier Science, Burlington, 6th ed. edition, 2013.

[3] Peter F. Jacobs Stuart G. Flechner and Richard T. Whitcomb. A high subsonic speed wind-tunnel investigation of winglets on a representative second-generation jet transport wing. Technical report, NASA, Langley Research Center, Hampton, Va. 23665, July 1976.

[4] Samuel Merryisha and Parvathy Rajendran. Review of winglets on tip vortex, drag and airfoil geometry. 2019. URL `https://api.semanticscholar.org/CorpusID:251620616`.

[5] Snorri Gudmundsson. *General Aviation Aircraft Design: Applied Methods And Procedures*. Butterworth-Heinemann is an imprint of Elsevier, The Boulevard, Langford Lane, Kidlington, Oxford OX5 1GB, UK 225 Wyman Street, Waltham, MA 02451, USA, 2014.

[6] User's Guide Reference manual. *Tornado Manual*, 2001. URL `https://tornado.redhammer.se/images/manual.pdf`. Accessed: 2024-06-15.

[7] Jacob R. Weierman. Winglet design and optimization for uavs. Master's thesis, University Stillwater, OK, USA, 2003.

[8] Margason Lamar. Aerodynamics of 3 d lifting surfaces through vortex lattice methods. URL `https://api.semanticscholar.org/CorpusID:50484285`. Accessed: 2024-06-15.

[9] Joaquim Peiro. Applications of computational fluid dynamics. Computer Laboratory Guides, 2023-2024.

[10] PhD Harry Smith. Aircraft flight mechanics, range and endurance, 2022. URL `https://aircraftflightmechanics.com/AircraftPerformance/RangeandEndurance.html`. Accessed: 2024-06-15.

[11] Liang ZHANG, Dongli MA, Muqing YANG, and Shaoqi WANG. Optimization and analysis of winglet configuration for solar aircraft. *Chinese Journal of Aeronautics*, 33 (12):3238–3252, 2020. ISSN 1000-9361. doi: https://doi.org/10.1016/j.cja.2020.04.008. URL `https://www.sciencedirect.com/science/article/pii/S1000936120301783`.

[12] Daniel P. Raymer. *Aircraft design : a conceptual approach*. American Institute of Aeronautics and Astronautics, Inc, Reston, VA, 2018. ISBN 1624104908. Includes bibliographical references and index.; ID: alma991000248798801591.

[13] Stephen Liu. Halo reach: Wing loads and structural design. GDP report 02015258, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[14] Kent R. Crenshaw Captain USAF. *Stability And Control Characteristics Of The Winglet Configured KC-135A*. PhD thesis, United States Air Force Air University Air Force Institute Of Technology, 1977.

[15] Ruairidh Scott-Brown. Halo reach: Empennage and lateral control design. GDP report 01734080, Dept. of Aeronautical Engineering, Imperial College London, 2024.

[16] Keith Koenig. *Interferance effects on the drag of bluff bodies in tandem.* PhD thesis, California Institute of Technology, 1978.

Appendix

A Comparison between wind tunnel data and results from Tornado VLM

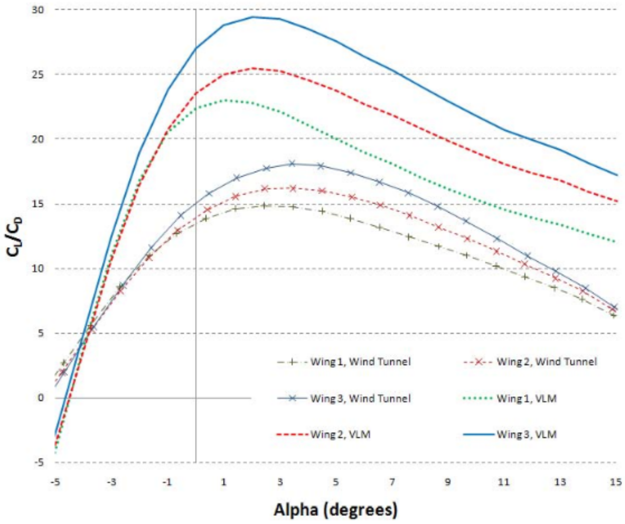


Figure 17. Wing 1 is the clean wing, wing 2 includes the lower winglets and a straight upper winglet. Wing 3 includes the lower winglets and blended upper winglet [7]

B Wing in Star CCM+

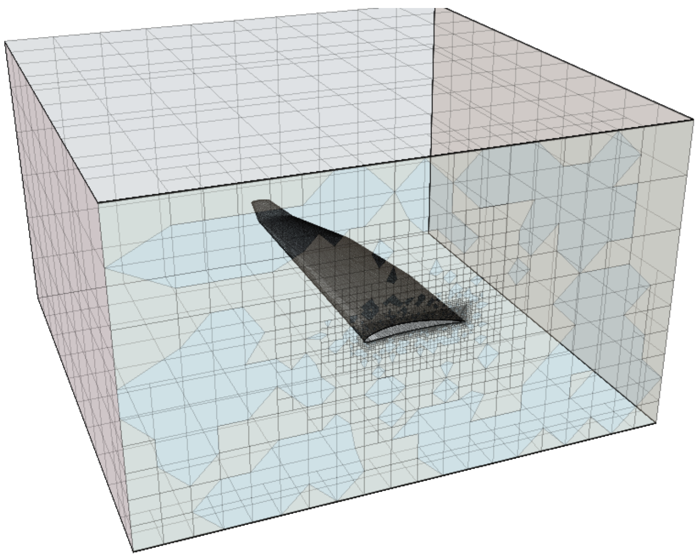


Figure 18. Half-wing set up in a virtual wind tunnel in Star CCM+

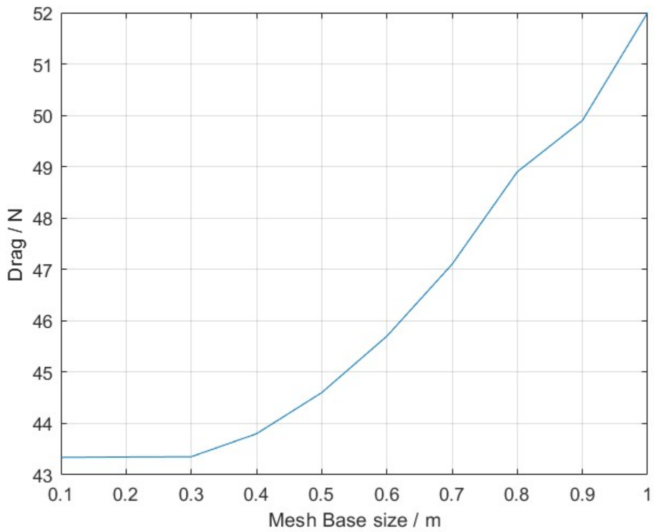


Figure 19. Drag on half of the wing with changing mesh size

C Horizontal Tailplane blanketing on Star-CCM+

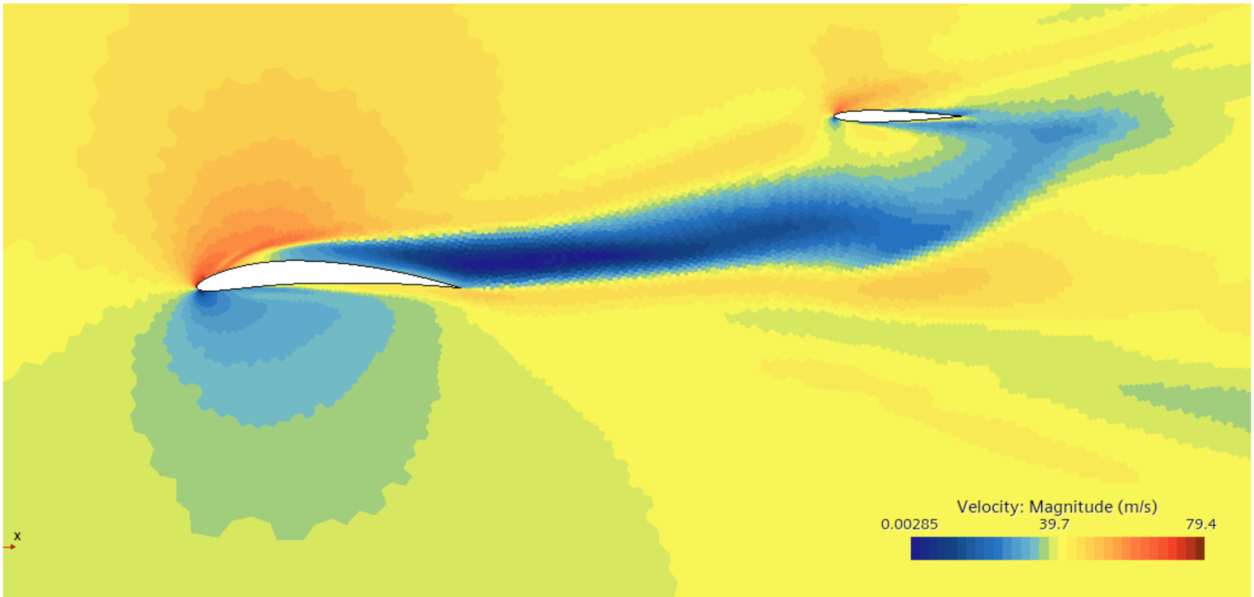


Figure 20. Blanketed Tail